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सत्यमेव जयते

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Standard Treatment Workflow (STW)

NEONATAL EMERGENCY TRIAGE ASSESSMENT AND MANAGEMENT

SICK OR AT-RISK NEONATE PRESENTING TO HEALTH FACILITY

- Place under radiant warmer
- Attach temperature probe and pulse oximeter
- Assess for emergency signs using TABC & CD

T: Temperature; **A**: Airway; **B**: Breathing; **C**: Circulation; **C**: Coma/convulsions; **D**: Dehydration

Classify into EMERGENCY, PRIORITY or NON-URGENT based on signs given below

EMERGENCY

- Significant hypothermia (<35.5°C)
- Apnea or gasping
- Respiratory distress (retractions, grunt)
- Central cyanosis or SpO_2 <91%
- Shock (cold peripheries, mottled/grey skin, CRT >3 s, weak pulses, low BP)
- Convulsions/coma
- Severe dehydration
- Persistent tachycardia >200/min

PRIORITY

- Temperature 35.5 to 36.4°C
- Weight <1800 g
- Large for gestation with risk factors (IDM)
- Tachypnea without retractions or grunt
- Severe jaundice (onset <24 h, palm or sole staining)
- Severe pallor
- Bleeding
- Major malformations
- Abdominal distension
- Irritable/restless, refusal to feed, decreased activity

NON-URGENT

- Mild jaundice
- Transitional stools
- Minor birth trauma
- Minor malformations
- Superficial infections
- Breastfeeding difficulty
- Regurgitation

TREAT IMMEDIATELY

- Maintain TABC
- Respiratory support: Nasal prong O_2 if SpO_2 <91% (titrate to 91–95%); nasal CPAP if respiratory distress; intubate if gasping or poor effort.
- Check blood glucose; send CBC and electrolytes
- If sepsis is likely, draw blood culture and give the first antibiotic dose without delay
- Start IV fluids as per weight and postnatal age
- Manage shock, dehydration and seizures

ASSESS PROMPTLY AND ACT RAPIDLY

- Maintain TABC
- Nasal prong oxygen if SpO_2 <91%
- Check blood glucose and act accordingly
- Start IV fluids (if abdominal distension/GI malformation) or gavage feeds
- Assess the risk factors for sepsis
- Investigations as per clinical findings

ASSESS AND COUNSEL

- Treat as per indication
- Counsel for breastfeeding
- Explain danger signs and follow-up

Follow specific STWs
A neonate may have more than one condition

SPECIFIC MANAGEMENT WORKFLOWS

SHOCK	• IV NS 10 mL/kg if volume expansion is indicated: preterm over 30–60 min; term over 15–20 min. • Consider inotropes early.
SEVERE DEHYDRATION **(Diarrhoea plus any two of lethargy, very slow skin pinch and sunken eyes)**	• Diarrhoeal severe dehydration: IV RL/NS 30 mL/kg in 1 h, then 70 mL/kg over 5 h (WHO Plan C). • If IV not possible: ORS 20 mL/kg/h for 6 h. Assess 1–2 hourly and titrate fluids. • Hypernatremic dehydration related to poor feeding/hot environment: manage as per the standard neonatal protocol.
SEIZURES **(Refer to STW on neonatal seizures)**	• Inj. phenobarbitone 20 mg/kg IV over 15 min. If seizures persist after 15 min, give a further 10 mg/kg IV over 10 min. • If uncontrolled, consider levetiracetam as second-line treatment. • Correct glucose and calcium if abnormal, concurrently.
SURGICAL	• Cover exposed skin defects with warm saline sterile gauze; maintain hydration. • **CDH:** Avoid CPAP/bag-mask ventilation. Intubate if necessary; decompress stomach with an OG/NG tube. • **Suspected TEF/oesophageal atresia:** Nil by mouth, IV fluids, head-up position; upper-pouch suction. • Arrange urgent referral to a surgical centre.
PERSISTENT HR >200/MIN	• Check ECG for arrhythmia. • Check and correct hyperthermia; refer for further management.
HYPOTHERMIA **(Refer to STW on thermal care of newborn)**	• Temperature <35.5°C: nurse under a radiant warmer in servo mode. • Temperature 35.5–36.4°C and baby is well: warm environment, skin-to-skin contact; recheck in 30 min. • If unwell or temperature fails to improve, nurse under a warmer.
HYPOGLYCEMIA **(Refer to STW on neonatal hypoglycemia)**	• Blood glucose <45 mg/dL and asymptomatic: supervised breastfeeding or EBM. • Blood glucose <20 mg/dL OR symptomatic: 2 mL/kg 10% dextrose IV, followed by infusion at 6 mg/kg/min.
JAUNDICE **(Refer to STW on neonatal jaundice)**	• Serious jaundice (onset <24 h, palms/soles yellow, or signs of acute bilirubin encephalopathy): intensive phototherapy; consider IV fluids if dehydrated; prepare for exchange transfusion as indicated.

DOs

- Screen every neonate immediately on arrival for emergency and priority signs.
- Maintain TABC, assess SpO_2 and blood glucose, and start indicated treatment promptly. Reassess after each intervention.
- Administer the first dose of antibiotic within 1 hour if there is a high likelihood of sepsis.

DON'Ts

- Do NOT delay emergency treatment for investigations or documentation.
- Obtain blood culture before the first dose of antibiotics, but do not delay treatment.
- Do NOT discharge a neonate with persistent emergency or priority signs; arrange admission or appropriate referral.

KEY PERFORMANCE INDICATORS (KPIs)

Triage classification documented (%)

$$\frac{\text{Neonates with a documented triage category at initial assessment}}{\text{All neonates assessed}} \times 100$$

Complete triage measurements (%)

$$\frac{\text{Sick neonates with temperature, } SpO_2 \text{ AND blood glucose recorded at initial assessment}}{\text{All sick neonates assessed}} \times 100$$

TRIAGE EVERY NEONATE ON ARRIVAL — TREAT EMERGENCY SIGNS IMMEDIATELY AND ASSESS PRIORITY SIGNS PROMPTLY

ABBREVIATIONS

BP: Blood pressure
CBC: Complete blood count
CDH: Congenital diaphragmatic hernia
CPAP: Continuous positive airway pressure
CRT: Capillary refill time
EBM: Expressed breastmilk
HR: Heart rate
IDM: Infant of a diabetic mother
NS: Normal saline
OG/NG: Orogastric/nasogastric
ORS: Oral rehydration solution
SpO_2: Oxygen saturation
TEF: Tracheoesophageal fistula

REFERENCES

- FBNC: MoHFW. Facility Based Newborn Care: Training Module for Doctors and Nurses (2023).
- WHO: Paediatric Emergency Triage, Assessment and Treatment (ETAT): Updated guideline (2016).

This STW has been prepared by national experts of India with feasibility considerations for various levels of healthcare system in the country. These broad guidelines are advisory, and are based on expert opinions and available scientific evidence. There may be variations in the management of an individual patient based on his/her specific condition, as decided by the treating physician. There will be no indemnity for direct or indirect consequences. Kindly visit the website of DHR for more information: (**stw.icmr.org.in**) for more information.
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